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Advancing predictive modeling in computational chemistry through quantum chemistry, molecular mechanics, and machine learning

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Abstract

Computational chemistry plays a critical role in advancing molecular science by bridging theoretical frameworks and experimental observations. It provides detailed insight into the structural, electronic, and reactive properties of molecules and materials. This article examines recent developments that are influencing the direction of the field, with a focus on the integration of quantum chemistry (QC), molecular mechanics (MM), and machine learning (ML) into cohesive modeling strategies. The objective is to assess how these combined approaches are improving the accuracy of simulations, informing molecular design, and contributing to progress in areas such as drug discovery, catalysis, and materials engineering. The review covers core methodologies including ab initio quantum mechanical calculations, hybrid quantum mechanics/molecular mechanics (QM/MM) models, and classical MD techniques. It also discusses emerging advances in data-driven models and neural network-based potentials. More attention is given to recent studies (2018–2025) that demonstrate how these techniques complement one another in addressing complex chemical systems. The analysis shows that combining quantum methods with ML enhances electronic structure predictions, while molecular mechanics provides efficient, large-scale modeling of structural and energetic properties across diverse environments, especially when coupled with simulation techniques such as molecular dynamics or Monte Carlo. In synergy, these tools support the construction of more robust and scalable models, narrowing the gap between computational results and laboratory findings. The article concludes by outlining the expanding influence of integrated computational approaches and their potential to drive innovation across scientific disciplines.

Keywords Computational chemistry, Quantum chemistry, Molecular mechanics, Machine learning, Predictive modeling

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1 Introduction

Computational chemistry is a rapidly evolving discipline that employs theoretical frameworks and computer-based simulations to investigate the structural, electronic, and reactive characteristics of molecules and materials [35]. Positioned at the intersection of experimental chemistry and theoretical physics, it delivers atom-level insights critical to advancements in catalysis, drug design, and materials science [138]. Its applications span the elucidation of reaction mechanisms, prediction of molecular behavior, and the rational design of functional compounds and materials [140, 192].

The foundation of the field lies in quantum mechanical techniques, particularly ab initio methods that derive molecular properties directly from physical principles, independent of empirical data [61]. Among these, Hartree–Fock theory and its post-Hartree–Fock extensions established a rigorous basis for electronic structure determination, though their high computational demands often constrained their use [68]. The advent of density functional theory (DFT) provided a more computationally feasible alternative, offering a favorable balance between accuracy and efficiency that broadened the accessibility of quantum chemical modeling [72]. Further methodological innovations, such as hybrid Quantum Mechanics/Molecular Mechanics (QM/MM) schemes, have enabled the simulation of complex environments, including biomolecular systems and solvated phases, where interactions like hydrogen bonding and van der Waals forces are pivotal [38], (Sousa et al., 2020).

While predictive modeling has long been a central goal of computational chemistry, ongoing advances are enhancing its accuracy and applicability. The integration of machine learning (ML) and artificial intelligence (AI) has enabled the development of data-driven tools capable of identifying molecular features correlated with target properties, thereby accelerating discovery while minimizing reliance on trial-and-error experimentation [70, 85, 173]. These models now contribute significantly to the accurate prediction of binding affinities, reactivity profiles, and material performance [153, 167].

It is important to distinguish between the models used to describe molecular interactions and the simulation techniques used to extract information from those models. Quantum chemistry, molecular mechanics, and machine learning–based potentials all represent different levels of detail in how the potential energy surface is defined and forces are computed. In contrast, simulation techniques propagate atomic motions according to Newton's or Hamiltonian mechanics and can be applied on top of any of these models, whether the forces are derived from classical force fields, quantum chemical calculations, or machine-learning interatomic potentials [20, 44].

This review explores recent breakthroughs and the evolving landscape of computational chemistry, focusing on the interplay between QC, MM, and ML. It begins by examining innovations in quantum methods that enhance electronic structure accuracy, followed by the role of simulation techniques in modeling conformational flexibility and thermally driven processes. A synergy of these developments is reshaping the methodological framework of the field, fostering more precise, efficient, and scalable strategies for tackling complex chemical problems.

2 Advances in quantum chemistry

2.1 Overview and advances of quantum chemistry methods

Quantum chemistry serves as the theoretical bedrock of computational chemistry, offering a rigorous framework for understanding molecular structure, reactivity, and properties at the atomic level [35, 180]. Over the last decade, the field has undergone substantial refinement in both methodological accuracy and scalability, enabling the investigation of systems of increasing complexity [77, 84]. Key computational strategies include the Hartree–Fock (HF) method, Density Functional Theory (DFT), and a range of post-Hartree–Fock approaches, each characterized by distinct trade-offs between computational efficiency and precision [15, 78].

Hartree–Fock, one of the earliest quantum chemical models, approximates electrons as independent particles moving in an averaged electrostatic field produced by other electrons [54]. Although HF is widely used as a reference for more sophisticated techniques, its failure to account for electron correlation limits its predictive accuracy, particularly for interaction energies and bond dissociation [12, 175–177].

Density Functional Theory improves upon HF by shifting the focus from wavefunctions to electron density, thereby reducing computational demands while incorporating electron correlation through exchange–correlation functionals [52, 169]. This balance of cost and accuracy has led to DFT's widespread use in calculating ground-state properties of medium to large molecular systems [22]. However, its reliability is contingent on the functional employed and may diminish in cases involving strong correlation, dispersion interactions, or complex transition structures [69, 96].

Enhancements to DFT have addressed many of these shortcomings. Notable improvements include the development of range-separated and double-hybrid functionals, as well as empirical dispersion corrections such as DFT-D3 and DFT-D4 [76, 168]. These refinements have extended DFT's applicability to non-covalent systems, transition states, and electronically excited configurations relevant to catalysis, photochemistry, and materials design [23, 67].

In addition to ab initio and density functional approaches, semiempirical quantum chemical methods have also advanced considerably. Recent developments such as GFN2-xTB [11] offer broad applicability with significantly reduced computational cost, making them valuable for large-scale screening and geometry optimization. Moreover, the integration of semiempirical quantum methods with ML has gained traction, enabling hybrid models that leverage physics-based approximations and data-driven corrections [121]. These methods expand the toolkit available for balancing accuracy and efficiency in contemporary QC.

Post-Hartree–Fock methods, including Møller–Plesset perturbation theory (MP2), Configuration Interaction (CI), and Coupled Cluster (CC) theory, address electron correlation directly and offer greater accuracy for a variety of molecular properties [59, 175–177]. Among these, the Coupled Cluster with Single, Double, and perturbative Triple excitations (CCSD(T)) method is widely regarded as the benchmark for precision in quantum chemistry [56, 130]. Nonetheless, the computational cost of these methods, which increase steeply with system size, typically confines their routine use to small or medium-sized molecules [92, 145].

Advances in quantum computing are opening new possibilities for chemical modeling. Algorithms such as the Variational Quantum Eigensolver (VQE) and Quantum Phase

Estimation (QPE) are being developed to address electronic structure problems more efficiently than is possible with classical computing [24, 155]. Although current implementations are limited by qubit instability and hardware noise, ongoing developments in error correction and device architecture are gradually making it feasible to simulate strongly correlated systems [8, 112]. Initial quantum simulations of simple molecules, including H_2, LiH, and BeH_2, highlight the promise of these methods for future applications [97].

Complementing these computational advances, fragment-based and multi-scale quantum mechanical techniques offer practical strategies for handling large systems. Methods such as the Fragment Molecular Orbital (FMO) approach, ONIOM (Our own N-layered Integrated molecular Orbital and molecular Mechanics), and the Effective Fragment Potential (EFP) model enable localized quantum treatments of subsystems within broader classical environments [93, 158]. These frameworks have proven especially useful in modeling enzymatic reactions, ligand binding, and solvation phenomena, where both quantum detail and large-scale context are essential [9, 48, 85].

These methodological innovations are reshaping the landscape of QC. Improvements in electron structure theory, the integration of quantum computing technologies, and the adoption of scalable modeling approaches are enhancing the reliability and scope of computational predictions across diverse areas of chemical research.

2.2 Applications

Quantum chemistry has undergone substantial development over recent decades, evolving from simple approximations to sophisticated computational methods capable of accurately modeling complex molecular systems [35]. This advancement has been driven by both enhanced computational resources and improvements in algorithms, establishing QC as a fundamental tool for predictive modeling within molecular sciences [138].

Significant advances have been made in understanding and predicting chemical reaction mechanisms. Quantum chemical methods such as DFT, coupled-cluster techniques, and multi-configurational approaches now enable detailed mapping of potential energy surfaces and characterization of transition states with high accuracy [40, 43]. Beyond these traditional approaches, recent years have seen rapid progress in the automated exploration of reaction pathways**,** where algorithms systematically generate and evaluate possible intermediates and transition states without requiring manual intuition [110, 142]. This has given rise to the broader field of chemical reaction network (CRN) analysis**,** which integrates high-throughput quantum chemistry with graph-based or machine learning methods to identify kinetically relevant pathways within complex networks [41, 63, 108, 151]. Studies have highlighted how automated mechanism discovery and CRN modeling are reshaping our ability to connect molecular reactivity to macroscopic kinetics, thereby offering new opportunities for catalyst design, materials discovery, and systems chemistry [110, 159, 165].

These techniques facilitate the identification of transition states, intermediates, and activation energies, providing molecular-level insights into chemical reactivity previously unattainable. Consequently, QC plays a crucial role in the rational design of catalysts and the optimization of reactions in both research and industrial contexts [100].

Quantum chemical methods also excel in determining the electronic structure of complex molecules. Applications range from organometallic catalysts and coordination

compounds to extended π-systems and bioinorganic clusters, where these methods reveal details about electron density distribution, oxidation states, and bonding characteristics [119]. Such insights are especially valuable in systems that challenge classical force fields due to significant electronic delocalization or correlation. Recent advancements in hybrid functionals, localized orbital methods, and embedding techniques have broadened the applicability of QC to larger and more chemically realistic systems [13, 118].

The study of photochemistry and excited states has similarly benefited from these advances. Techniques including time-dependent DFT (TD-DFT), complete active space self-consistent field (CASSCF), and equation-of-motion coupled-cluster (EOM-CC) approaches offer detailed understanding of light-induced phenomena, electronic excitations, and relaxation processes [60, 122, 170]. These capabilities are central to the development of materials for applications such as photovoltaics, photodynamic therapy, and molecular electronics. Ongoing improvements in accuracy and computational efficiency continue to deepen our understanding of molecular interactions with light [36].

These advances in QC have significantly expanded the scope of computational studies, enabling precise prediction and control of molecular behavior. They remain foundational to the integration of data-driven approaches and multi-scale simulations in the broader field of computational chemistry [16, 120].

3 Molecular mechanics

3.1 Overview, recent advances in, and applications of molecular mechanics

Molecular mechanics (MM) is a cornerstone of computational chemistry, offering a framework to model molecular systems by treating atoms as classical particles and interactions through empirical or semi-empirical potential energy functions known as force fields [94]. Unlike quantum chemical approaches, which explicitly solve the electronic Schrödinger equation, MM approximates the potential energy surface through parametrized functions that describe bonded and non-bonded interactions. This allows MM to handle very large systems, ranging from biomolecules to materials, at a fraction of the computational cost of quantum mechanical methods, making it indispensable in multi-scale modeling [33, 80].

Over the decades, MM has evolved into diverse branches, each designed to address specific limitations of classical models and to extend applicability across scales and chemical environments.

3.2 Classical (All-Atom) molecular mechanics

Classical or all-atom molecular mechanics (MM) explicitly represents every atom in a system, with the total potential energy expressed as a sum of bonded interactions, including bond stretching, angle bending, and torsional rotations, as well as non-bonded terms such as electrostatics and van der Waals forces [64]. Widely used force fields such as AMBER, CHARMM, OPLS-AA, and GROMOS have been extensively validated and optimized for biomolecular simulations, making them standard tools in computational chemistry [154]. These all-atom models have proven highly effective for studying proteins, nucleic acids, membranes, small molecules, and drug design, where atomistic detail is essential for capturing structural and dynamic properties. Recent advances in this field have focused on improving the transferability of force fields across broader

regions of chemical space, refining the treatment of challenging systems such as metal centers, and integrating GPU acceleration into MM engines to achieve significant gains in efficiency [82, 98]. Furthermore, the combination of MM with enhanced sampling techniques and hybrid QM/MM frameworks has improved the accuracy of simulations, particularly for enzymatic reactions and ligand-binding studies, where both large-scale dynamics and localized electronic effects must be accurately captured [175–177].

3.3 Coarse-grained (CG) molecular mechanics

Coarse-grained (CG) models simplify molecular systems by reducing complexity, grouping several atoms into larger interaction centers, or "beads." This abstraction dramatically lowers computational cost, making it possible to simulate much larger systems over biologically relevant timescales than would be feasible with fully atomistic models [117]. Widely adopted frameworks such as MARTINI, SDK, and UNRES have enabled CG simulations to play a central role in studying membranes, protein assemblies, viral capsids, and a variety of mesoscale biological processes [104, 144]. In recent years, CG force fields have undergone systematic refinement to enhance both structural fidelity and dynamic accuracy, ensuring that essential biophysical properties are preserved despite the reduced resolution. A particularly significant advance has been the integration of machine learning into CG parameterization, which allows automated mapping from all-atom simulations and the development of more transferable CG models [102]. Additionally, hybrid CG/all-atom approaches are gaining prominence, where CG models provide efficient low-resolution sampling, followed by backmapping procedures to recover atomic detail for higher-resolution refinement. These advances are reshaping coarse-grained modeling into a more precise, adaptable, and data-driven approach for multiscale simulations of complex biological systems [128, 133].

3.4 Polarizable force fields

Traditional molecular mechanics (MM) force fields rely on fixed atomic charges, which limit their ability to adapt to changes in electrostatic environments. To address this, polarizable force fields have been developed, allowing electron density redistribution through mechanisms such as induced dipoles or Drude oscillators. Notable examples include AMOEBA and Drude-CHARMM, which provide more realistic modeling of ion solvation, protein–ligand recognition, and interfacial chemistry [81]. These improvements extend the applicability of MM to biologically and chemically significant problems, including protein–ligand binding, nucleic acid stability, solvation studies, and membrane interactions [5, 194]. Recent advances have focused on creating next-generation polarizable models with refined parameterization pipelines, enhanced scalability for large systems, and GPU-enabled implementations. Polarizable force fields are becoming increasingly practical for routine biomolecular simulations, marking a significant advance toward more accurate and transferable computational frameworks [99].

3.5 Reactive force fields

A key limitation of classical MM is its inability to capture chemical reactivity, since bonds are treated as permanently fixed and cannot form or break during a simulation. Reactive force fields (RFFs) were developed to overcome this constraint by introducing a flexible framework in which bonds can dynamically rearrange as a function of interatomic

distances and local environments [129]. These models are trained on quantum chemical datasets, ensuring that the underlying parameters retain a close connection to high-level electronic structure information while still being computationally tractable for large-scale simulations. Among the most widely used frameworks are ReaxFF, which has become a general-purpose tool for modeling reactive processes; AIREBO**,** which is particularly effective for describing carbon-based systems such as graphene and nanotubes; and COMB, which has been tailored for ceramics, oxides, and metallic systems [107, 184].

The ability of RFFs to model bond dynamics has opened broad application domains, including combustion chemistry, heterogeneous catalysis, electrochemical processes, battery chemistry, and nanomaterials. For instance, ReaxFF is routinely applied to study flame propagation and hydrocarbon oxidation, while COMB has proven valuable in investigating ionic conductivity and stability in solid-state electrolytes [30]. Recent advances have further enhanced the accuracy and predictive power of RFFs. Automated parameterization workflows now enable faster and more systematic development of force fields by leveraging quantum chemistry databases, while ongoing improvements focus on extending these models to handle increasingly complex interfacial environments. This progress is particularly relevant for solid–liquid and solid–electrolyte boundaries**,** which play central roles in energy storage technologies and catalytic transformations [90, 163]. Reactive force fields are transforming molecular mechanics from a static representation of molecular structures into a dynamic and chemically responsive framework, one that bridges the gap between quantum chemistry and large-scale simulations.

3.6 Specialized and emerging approaches

Beyond the classical, coarse-grained, polarizable, and reactive formulations, several specialized branches of MM have emerged to tackle unique challenges in modeling complex chemical and biological systems. The empirical valence bond (EVB) method, for instance, is particularly effective in simulating reactive processes by interpolating between predefined reactant and product states, making it a powerful tool in the study of enzyme catalysis [2].

Similarly, structure-based force fields have been developed as simplified models tailored for protein folding and large-scale structural transitions, providing a balance between computational efficiency and the ability to capture essential conformational dynamics. More recently, the integration of machine learning into MM frameworks has created hybrid approaches that refine classical force fields by learning corrections from high-level quantum chemical data or, in some cases, by generating fully data-driven potentials. These ML-augmented force fields represent a significant advance, serving as a bridge between traditional MM models and modern machine learning–based interatomic potentials, thereby improving both accuracy and transferability across chemical systems [102, 120].

Molecular mechanics has diversified from its classical origins into a hierarchical family of methods, all-atom, coarse-grained, polarizable, reactive, and ML-augmented, each tailored to balance accuracy, efficiency, and scale. The field is undergoing a paradigm shift driven by ML integration, GPU acceleration, and multiscale modeling frameworks, making MM more versatile and predictive than ever. Its role as a complement to QC and

ML ensures that MM remains central to modern computational chemistry, enabling the study of systems that span molecular, biological, and material scales.

4 Machine learning in computational chemistry

4.1 Role of data-driven models

The integration of ML into computational chemistry has ushered in a paradigm shift from purely physics-based simulations to data-driven predictions. Machine learning models are increasingly being employed to accelerate chemical discovery, reduce the computational cost of quantum mechanical calculations, and uncover patterns in large and complex molecular datasets [120, 172]. At the core of this transformation is the ability of ML algorithms to learn from data and make accurate predictions about molecular properties, reactivities, and behaviors without explicitly solving the underlying physical equations for each new instance [136].

Machine Learning models are generally categorized into supervised learning, unsupervised learning, deep learning and more recently self-supervised learning (SSL) techniques [53]. Each category serves distinct purposes and offers unique advantages in analyzing molecular data and predicting chemical properties.

Supervised learning is the most prevalent method and involves developing models using datasets where each molecular structure is paired with known properties such as solubility, reactivity, or binding affinity. By training on this labeled data, these models learn the relationship between molecular features and their properties, enabling them to predict the characteristics of new, untested molecules with considerable accuracy [51]. This approach has found wide application in drug discovery, material design, and property prediction, where accurate forecasting of molecular behavior is essential.

Unsupervised learning, in contrast, operates without labeled datasets. Instead, it focuses on uncovering underlying structures or patterns within molecular collections. This approach is especially valuable for clustering molecules with similar chemical traits or simplifying complex, high-dimensional descriptor spaces through dimensionality reduction techniques. By revealing intrinsic relationships, unsupervised learning aids in identifying novel compound classes and organizing chemical space more effectively [3, 50, 106].

Deep learning, a sophisticated extension within supervised learning, utilizes neural networks composed of multiple layers that can automatically extract and learn relevant features directly from raw molecular data [174, 179]. Advanced neural network architectures have demonstrated remarkable capabilities in computational chemistry tasks that were once considered highly challenging. They include the Graph neural networks (GNNs), Convolutional neural networks (CNNs), and the Recurrent neural networks (RNNs). GNNs are particularly well suited to molecular modeling because they operate directly on molecular graphs, capturing the relational information between atoms and bonds in a way that aligns naturally with chemical structure [28, 31]. CNNs, originally developed for image recognition, have been successfully adapted to chemical applications by treating molecular structures, electron density maps, or protein–ligand binding pockets as grid-based spatial data, allowing the extraction of local structural features relevant to binding and reactivity [55, 174, 179]. RNNs, and their modern variants such as long short-term memory (LSTM) and gated recurrent units (GRUs), have shown strength in handling sequential data representations such as SMILES strings, enabling

robust prediction of molecular properties, retrosynthetic pathways, and reaction outcomes [115, 189].

In recent years, self-supervised learning (SSL) has emerged as a powerful paradigm that bridges aspects of supervised and unsupervised learning. SSL leverages unlabeled data by generating surrogate pretext tasks, such as predicting masked tokens in SMILES strings or reconstructing molecular graphs, which allows models to learn chemically meaningful representations without the need for extensive labeled datasets [188, 191]. These pre-trained models can then be fine-tuned for downstream tasks such as property prediction, virtual screening, or reaction outcome modeling, often with improved performance in low-data regimes [103].

These architectures have pushed the boundaries of what is possible in computational chemistry, from planning synthetic routes to predicting reaction outcomes and screening vast chemical libraries. Each brings a different strength to the table, whether it's recognizing structural patterns in molecules, modeling the flow of reactions, or rapidly evaluating new compounds, and their combined use has made computational workflows more accurate, efficient, and insightful than ever before [45].

4.2 Input representations

The effectiveness of ML models in chemistry is closely tied to the way molecular structures are represented as input data. Accurate and meaningful representations are essential because they enable ML algorithms to interpret complex chemical information and make reliable predictions. Over the years, several molecular encoding methods have been developed to transform chemical structures into numerical formats suitable for computational analysis.

One of the most commonly used representations is the Simplified Molecular Input Line Entry System (SMILES). SMILES encodes a molecule as a linear string of characters that denote atoms and the bonds connecting them, following a defined sequence. Its popularity stems from its simplicity and compatibility with natural language processing techniques, which facilitates integration with various ML frameworks. However, SMILES representations can be sensitive to changes in the order of atoms, meaning that the same molecule can have multiple valid SMILES strings, which may affect model performance [83, 137]. Subsequent studies have demonstrated that canonicalization resolves most of these inconsistencies, while the inherent redundancy of non-canonical SMILES can actually be leveraged for data augmentation, improving model robustness [195].

Another widely employed approach involves molecular fingerprints. These are fixed-length binary or integer vectors that indicate the presence or absence of particular substructures or molecular fragments within a compound. Fingerprints such as Morgan fingerprints—which use circular neighborhoods around atoms to generate unique identifiers, and MACCS keys are extensively used in Quantitative Structure–Activity Relationship (QSAR) modeling and similarity searches. By abstracting complex molecular information into these compact vectors, fingerprints enable efficient comparison and analysis of large chemical datasets [65, 111].

Graph-based molecular representations have gained considerable attention in recent years. In this format, molecules are viewed as undirected graphs where atoms correspond to nodes and chemical bonds correspond to edges. This approach allows the use

of Graph Neural Networks (GNNs), which are capable of learning directly from molecular connectivity and topology without requiring handcrafted features. GNNs have demonstrated impressive performance and generalizability across various molecular property prediction tasks, positioning them as a state- of-the-art method in computational chemistry [88, 91, 185].

These diverse data-driven methods are transforming computational chemistry by harnessing the power of machine learning to extract meaningful patterns from both experimental measurements and theoretical simulations. Such models enable chemists to rapidly screen vast molecular libraries, optimize reaction conditions, and predict properties of novel materials with greater accuracy and speed than traditional approaches [143, 183]. As advances in algorithm development continue alongside the growth of comprehensive molecular datasets, machine learning is poised to become a cornerstone of future predictive modeling efforts in chemistry, fundamentally reshaping how molecular discovery and design are conducted [131, 164].

4.3 Key applications

ML is profoundly reshaping the field of computational chemistry by providing data-driven approaches that complement traditional physics-based techniques. With the rapid expansion of chemical data, from quantum mechanical computations and experimental measurements to detailed structural and physicochemical descriptors, ML has become an essential tool for identifying hidden patterns, constructing predictive models, and accelerating the pace of discovery [51, 186].

One of the most significant applications of ML in this domain is the prediction of molecular properties. By training on carefully curated datasets containing molecules with experimentally or theoretically determined properties, ML models can accurately forecast various attributes such as aqueous solubility, lipophilicity, binding affinities, and toxicity. These predictions are generated much faster than traditional quantum mechanical methods, facilitating large-scale virtual screening efforts and reducing the dependency on costly laboratory experiments [148, 190]. Methods such as random forests and support vector machines have been widely employed, but recent advances in GNNs have particularly excelled at capturing the intricate relationships between molecular structure and its properties [132, 187].

Machine learning plays a vital role in predicting reaction outcomes and designing retrosynthetic pathways. While early strategies relied heavily on rule-based algorithms and expert knowledge, these approaches are still being actively refined and have recently achieved strong performance in specific domains [166, 178]. At the same time, deep learning methods trained on large reaction databases have enabled substantial progress in yield prediction, selectivity assessment, and pathway design. Architectures such as the Molecular Transformer and subsequent attention-based or graph neural network models have reached impressive levels of accuracy for both forward and retrosynthetic prediction, even extending to chemistries not represented in their training sets [123, 171]. More recent benchmarks and open databases, such as the USPTO-MIT datasets and the Open Reaction Database, have further supported the systematic evaluation of these models [182]. Despite significant advances, many machine learning models still face challenges. They are often affected by biases in their training data, perform poorly with reaction types that have limited examples, and can be difficult to interpret or link

to underlying chemical mechanisms. Overcoming these hurdles will likely require blending the precision of rule-based and symbolic methods with the adaptability of data-driven learning to build hybrid models that are both accurate and rooted in chemical understanding.

Additionally, ML is revolutionizing de novo molecular design, where algorithms generate entirely new chemical structures tailored to meet specific requirements. Generative approaches, such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and reinforcement learning, can efficiently explore vast chemical spaces to design molecules that balance multiple desirable features, including biological activity, drug-likeness, and ease of synthesis. These strategies enable rapid generation of candidate compounds well before any experimental testing, thereby significantly shortening the drug discovery timeline [79, 193, 196].

As the discipline of computational chemistry continues to advance, the integration of machine learning marks a transformative shift. By converting vast quantities of chemical data into actionable predictions and insights, ML works synergistically with QC and MD methods. This convergence is poised to drive the next generation of predictive models, fostering more efficient, precise, and innovative approaches to chemical discovery and design [57, 120].

Beyond these established applications, ML is also driving innovation in other frontiers. Recent work has shown that ML can directly enhance sampling strategies in MDS, providing more efficient exploration of conformational landscapes [113]. In QC, ML models are increasingly coupled with density functional theory to improve accuracy and scalability [125]. Most recently, large language model–powered agents have been introduced to interact with and control computational chemistry software, marking a shift toward more autonomous simulation workflows [196].

5 Synergy between quantum chemistry, molecular mechanics, and machine learning

The integration of QC, MM, and ML is transforming computational chemistry by enabling detailed multi-scale modeling and enhancing predictive accuracy. Each methodology contributes distinct advantages: QC delivers precise electronic structure calculations; MM provides atom-level energy evaluations and equilibrium structures; and ML introduces powerful data-driven models capable of generalizing complex patterns. When combined, these approaches enable researchers to connect different spatial and temporal scales, achieving insights that were previously out of reach [16, 120].

Central to this progress is multi-scale modeling. QC excels at describing electronic interactions with high fidelity on femtosecond and nanometer scales but is limited by its computational demands when applied to large molecular systems or long simulation times [100]. Molecular mechanics complement this by providing efficient energy evaluations and equilibrium structures based on empirically derived force fields. When combined with molecular dynamics (MD), these force fields enable simulations of biomolecular behavior over microsecond timescales, though they lack the quantum mechanical precision of electronic-structure methods [126]. ML bridges these gaps by enabling the creation of ML-based force fields that approximate quantum mechanical accuracy while drastically reducing computational cost [146]. The resulting hybrid

models capture essential physical phenomena across varying degrees of complexity and scale [34].

5.1 Hybrid methods: quantum mechanical and molecular mechanical modeling (QM/MM)

A significant advancement in computational chemistry is the development of hybrid workflows that combine QC and MM methods, increasingly augmented by ML techniques. Classical QM/MM frameworks strike a balance between accuracy and efficiency by applying quantum chemical methods to critical regions, such as catalytic active sites, chromophores, or small ligands, while modeling the surrounding environment**,** such as proteins, membranes, or solvents, with MM force fields [6, 47]. This division enables highly accurate descriptions of localized electronic events while retaining computational tractability for large biological or material systems.

Recent progress has expanded the utility of QM/MM approaches in enzyme catalysis, drug binding, and reactive pathway elucidation**,** where electronic reorganization is localized yet strongly influenced by the surrounding environment. For instance, adaptive QM/MM techniques dynamically adjust the size and boundary of the QM region in response to the system's evolution, improving accuracy in highly flexible biomolecular contexts [37, 94].

ML has become a powerful complement to QM/MM by addressing two long-standing challenges: the high computational cost of quantum calculationsand theartificial boundary between QM and MM regions. Surrogate ML models, trained on high-level QC data, can approximate electronic energies and forces at near-DFT accuracy but with orders-of-magnitude faster evaluation times [37, 124]. Similarly, ML-driven boundary treatments can smooth the QM/MM interface, reducing artifacts associated with link atoms or charge redistribution. These innovations enable simulations of unprecedented scale, extending from femtosecond electronic transitions to millisecond enzymatic turnover [17, 175–177].

The ML-enhanced QM/MM workflows represent a paradigm shift in multiscale modeling**,** combining the accuracy of quantum mechanics**,** the scalability of MM, and the efficiency of ML**.** This integrated approach not only accelerates the study of enzyme catalysis and drug discovery but also opens new frontiers in modeling reactive pathways in complex condensed-phase and biomolecular systems [6, 87].

5.2 The role of molecular dynamics simulations

Molecular dynamics (MD) simulation has established itself as an essential technique in computational chemistry, providing detailed insights into the atomic-level, time-dependent behavior of molecular systems [152]. Unlike time-independent approaches, MD simulations capture the temporal evolution and dynamic behavior of molecules by numerically integrating Newton's equations of motion [4, 86].

At its core, MD treats atoms as classical particles whose trajectories are propagated through time by solving Newton's equations of motion (or Hamiltonian mechanics in practice). This propagation scheme is independent of how the underlying potential energy surface is represented, which makes MD broadly compatible with multiple modeling frameworks.

Historically, atomic forces were derived from molecular mechanics models known as force fields, which approximate bonded interactions, such as bond stretching, angle

bending, and torsional rotations, and non-bonded interactions such as van der Waals forces and electrostatics. Widely used force fields, including AMBER, CHARMM, and OPLS-AA, have been extensively validated for biomolecular simulations. While coarse-grained models have enhanced computational efficiency and enabled simulations of larger systems across longer timescales, polarizable models have improved accuracy in capturing electronic induction effects, albeit at a higher computational cost [26, 27, 126, 150, 154].

Crucially, MD is not limited to molecular mechanics. When coupled with QC methods, such as density functional theory, it enables ab initio *MD* in which forces are computed directly from electronic structure calculations. This provides electronic-level accuracy while retaining the temporal resolution of MD. When combined with MM force fields, MD becomes a workhorse for sampling conformational space efficiently and providing equilibrated structures for hybrid QM/MM approaches [94]. Finally, the integration of ML has opened new avenues, as MD can now be driven by machine learning–based interatomic potentials that combine near-quantum accuracy with classical efficiency. Thus, MD serves as a unifying simulation framework, augmenting QC with dynamics, extending MM beyond static structures, and providing a testing ground and training data for ML-based potentials [32, 127].

Despite these strengths, classical MD remains constrained by accessible timescales, often failing to capture rare but biologically significant processes such as protein folding, ligand binding, or large conformational transitions [18]. To address this, enhanced sampling strategies have been developed. Approaches such as metadynamics, umbrella sampling, and replica exchange molecular dynamics (REMD) enable systems to overcome energy barriers and access otherwise hidden states [71, 73, 89, 139]. For example, metadynamics applies a time-dependent bias to penalize revisited states, facilitating exploration of new conformations [75], while REMD employs multiple replicas under different conditions that periodically exchange configurations, enhancing conformational sampling [141, 156].

With advances in computational power and algorithms, the capability and precision of MD simulations have improved dramatically, enabling studies of increasingly complex systems ranging from enzymatic catalysis to drug–target interactions [1, 19, 162].

Cross-validation plays a vital role in ensuring the reliability of these integrated methods. ML models trained on quantum chemical data sets are often validated using molecular dynamics simulations to verify physical plausibility. Similarly, molecular dynamics outputs guide the selection of relevant molecular configurations for training ML models or for conducting high-level quantum calculations. This iterative feedback enhances the robustness and predictive power of the overall modeling approach [120].

Moreover, the success of combining QC, MM, and ML increasingly depends on collaborative data sharing and reproducibility standards. The field is moving toward open science principles, emphasizing the creation of standardized databases, interoperable software, and adherence to FAIR (Findable, Accessible, Interoperable, Reusable) data guidelines. Cloud computing platforms and shared repositories facilitate the dissemination of models, datasets, and workflows, thus accelerating research and ensuring reproducibility [39, 58].

Ultimately, the convergence of these three fields marks more than just a technical advancement; it represents a paradigm shift in computational chemistry toward a

comprehensive, multi-scale, and data-driven approach capable of tackling complex problems in chemistry, biology, and materials science [25].

6 Challenges and limitations

Despite significant progress in computational chemistry, several persistent challenges continue to impede the broader adoption, scalability, and dependability of predictive models based on QC, MM, and ML [120, 147]. A primary obstacle lies in the quality and accessibility of data. The development of reliable ML models and the validation of simulation outcomes depend heavily on high-quality experimental measurements and well-curated computational datasets. Unfortunately, many available datasets suffer from inconsistencies, incomplete metadata, and reproducibility issues, which undermine their utility for model training and evaluation [30, 161]. This problem is particularly acute in specialized chemical domains or emerging materials, where data scarcity can significantly limit model development and validation efforts [7].

Another fundamental limitation is the inherent trade-off between computational cost and accuracy. High-accuracy quantum mechanical methods, such as coupled cluster or multi-reference techniques, offer exceptional precision but demand substantial computational resources and scale poorly with increasing system size [15, 149]. Conversely, approximate methods or empirical force fields provide faster computations but often sacrifice predictive accuracy and reliability [181]. Achieving an optimal balance between computational feasibility and chemical fidelity remains a central challenge, especially when dealing with complex or large molecular systems [16].

Within ML applications, issues of model interpretability and generalizability persist. Many deep learning models operate as opaque systems, offering little transparency regarding the chemical principles underpinning their predictions [66, 105, 160]. Additionally, models trained on narrow molecular, or reaction datasets frequently struggle to generalize effectively to novel chemical spaces, limiting their practical applicability across diverse scenarios [190]. Developing ML frameworks that are both interpretable and robust across a broad range of chemical domains is crucial for fostering trust and enabling widespread adoption in real-world applications [190].

Software interoperability and standardization present further challenges to progress in computational chemistry. The diversity of computational platforms, each with its own file formats, user interfaces, and nomenclature, complicates data exchange and integration of workflows [42].

One persistent challenge in computational chemistry is the absence of broadly adopted protocols for inputs, outputs, and benchmarking, which undermines reproducibility and makes it difficult to compare methods consistently [21]. However, recent community-driven efforts have begun to address these issues. Benchmarking platforms such as Matbench provide standardized datasets and evaluation protocols for materials property prediction [10]. The Quantum Chemistry Metadata Language (QCML) offers a structured framework for representing quantum chemistry results in a machine-readable way [46]. In addition, the Open Reaction Database (ORD) has been established as a FAIR, extensible repository for reaction data, enabling more systematic benchmarking and data-driven method development [114]. These initiatives mark significant steps toward reproducible standards and robust benchmarks in computational chemistry. Addressing

the remaining gaps will be crucial for fostering collaboration, ensuring reproducibility, and accelerating progress in the field.

Addressing these multifaceted challenges will require coordinated community efforts focused on rigorous benchmarking protocols, open data initiatives, the design of transparent and interpretable ML models, and the establishment of unified software standards [74, 186]. Through such collaborative endeavors, computational chemistry can continue evolving into a more predictive and reliable discipline capable of addressing complex scientific problems.

7 Future directions

The field of computational chemistry is undergoing a profound transformation, driven by the integration of several emerging technologies that promise to greatly enhance the speed, precision, and breadth of predictive modeling. This evolution is set to reshape the discipline in fundamental ways over the coming years.

Foremost among these advances is the development of quantum computing and next-generation quantum algorithms. Unlike classical computers, quantum machines inherently handle complex quantum phenomena, making them particularly well-suited to solving the electronic Schrödinger equation for large, strongly correlated systems. As quantum hardware continues to improve in stability and error correction, QC is poised to experience dramatic increases in computational efficiency. This will enable more accurate simulations of reaction mechanisms, excited electronic states, and the intricate electronic structures of transition metal complexes [8, 24].

Simultaneously, artificial intelligence-driven autonomous discovery platforms are revolutionizing molecular and materials design. These platforms combine sophisticated computational models with high-throughput virtual screening and automated experimental systems, forming closed-loop discovery workflows. By minimizing human intervention, these integrated systems accelerate the generation and testing of hypotheses, streamline synthetic route optimization, and reduce both the time and cost associated with developing new drugs and catalysts [51, 95, 101, 124, 157].

Another significant advance is found in the area of molecular dynamics (MD) simulations enhanced through MLpotentials. Traditional force fields, while efficient, often fail to capture the nuanced accuracy required for reactive or unconventional chemical systems. Ab initio MDprovidehigh accuracy but at a substantial computational cost. Machine learning approaches, particularly neural network potentials trained on high-quality quantum mechanical data, can now achieve near-DFT accuracy at a fraction of the computational cost.

This breakthrough significantly extends the scope of atomistic simulations by enabling more accurate and efficient modeling of chemical and biological systems. However, it does not fully resolve the intrinsic timescale limitations of molecular dynamics, as rare reactive events and complex biological processes often remain inaccessible without employing enhanced sampling or biasing strategies [14, 116].

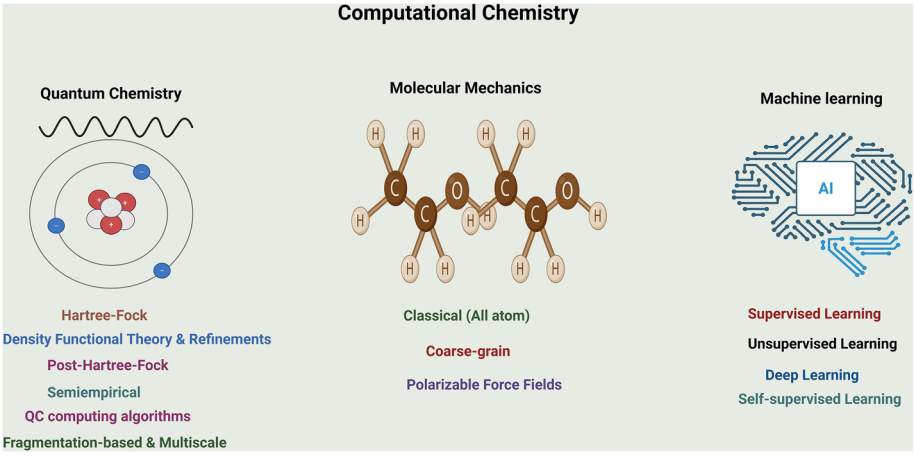
The concept of digital twins, dynamic, data-driven virtual representations of real-world chemical systems, is also emerging as a transformative tool. These models continuously assimilate both experimental observations and computational predictions to simulate system behavior under varying environmental conditions. While digital twins have long been applied in engineering, their use is expanding into chemical catalysis,

material degradation studies, battery development, and drug delivery systems. This approach enables adaptive system control and predictive diagnostics, offering powerful new avenues for research and industrial applications [129].

Finally, the future trajectory of computational chemistry will be shaped by open science principles and collaborative frameworks. Recent advances in open-source infrastructures, such as the Atomic Simulation Recipes [49] and AiiDA [62], have embedded FAIR data management directly into simulation workflows, ensuring provenance, reproducibility, and large-scale automation. In parallel, community databases—including the Materials Project [134], NOMAD [135], and the Open Catalyst Project [29], have dramatically expanded access to millions of quantum-mechanical calculations across chemical and materials spaces. These repositories are now central resources for training next-generation machine learning interatomic potentials such as NequIP [14] and Allegro [116], which approach DFT accuracy at a fraction of the computational cost.

The broader adoption of FAIR (Findable, Accessible, Interoperable, and Reusable) data standards, coupled with the growth of community-driven benchmarks and open-source software, will further enhance transparency, reproducibility, and cross-disciplinary innovation. Such practices are vital for developing robust, transferable models while democratizing access to state- of-the-art computational tools and datasets worldwide [109].

These trends herald a future where computational chemistry evolves into a more predictive, autonomous, and interdisciplinary field, capable of addressing some of the most complex and pressing problems across chemistry, biology, and materials science.



8 Conclusion

The landscape of computational chemistry is undergoing a profound transformation. The integration of QC, MM, and ML is enabling the development of predictive, interpretable, and scalable models for molecular systems. QC continues to evolve in accuracy and breadth, providing electronic-level descriptions of molecules and reactions. MM complements this by enabling efficient modeling of large systems and energy landscapes, while MD, built on MM or alternative potentials, offers time-resolved insights into system behavior. ML bridges the limitations of physics-based models by accelerating simulations and uncovering hidden patterns in complex chemical data. However, challenges remain. Data quality, model interpretability, and software interoperability must be addressed to ensure the reliability and reproducibility of predictive models.

Moreover, ethical concerns such as bias in ML training sets and unequal access to high-performance computing should not be overlooked.

Looking forward, the convergence of physics-based and data-driven methods is poised to reshape how we conduct chemical discoveries. Embracing FAIR data practices, fostering interdisciplinary collaboration, and investing in open science infrastructure will be crucial to realizing the full potential of computational chemistry as a driver of innovation in science and technology.

Supplementary Information

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Supplementary file 1.

Author contributions

ABR conceptualized the study, led the manuscript writing, and contributed to the sections on quantum chemistry and integrative modeling strategies. AAF conducted a comprehensive literature review and contributed to the section on molecular dynamics simulations and applications. TA assisted in data interpretation and drafted the section on machine learning approaches in predictive modeling. JCO contributed to editing, referencing, and synthesis of methodological trends across computational disciplines. OIR also did literature review and assisted in harmonizing the manuscript structure. TRF reviewed recent applications for AI in chemistry and contributed to the future perspectives section. DB critically reviewed the manuscript and provided key insights on translational applications in drug discovery. All authors reviewed and approved the final version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable. This study did not involve human participants or animals.

Consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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